		$\frac{1}{4}mv^2 - \frac{1}{2}mv^2 = -\frac{1}{4}mv^2 = -\frac{E_k}{2}$
5	**C**	Acceleration along the slope = $\frac{30-0}{6} = 5\ \mathrm{m\ s^{-1}}$. Resolve weight along and perpendicular to slope. Along the slope the component of weight is $mgsin\theta$. Hence Newton's 2nd law gives $m(gsin\theta) = 5m$, so $\theta = \sin^{-1}\left(\frac{5}{9.81}\right) = 30.6^o$
6	C	Let W be the weight on the board